DOCKHUB · CONCEPT OPTIONS

DockHub concept options

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DOCUMENT CONTROL

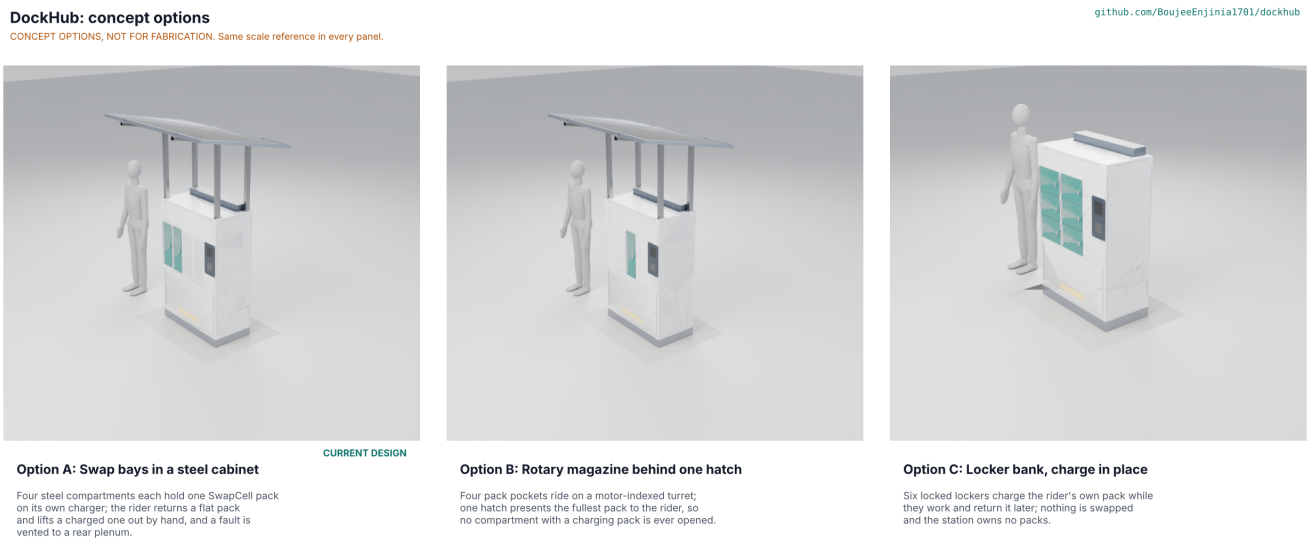
Document ID	DKH-OPT-001	Version	0.1
Status	Draft	Date	2026-10-31
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/dockhub		
Source	docs/02a-concept-options.md @ 5167e03		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-31	Amish Chadha	Three options with different working principles scored against DKH-URS-001 at the kit 1.9.1 upgrade; option A, the current design, is unchanged; the choice is proposed, awaiting Amish; every option modelled to D3 in cad/src/model.py

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Result: option A, swap bays in a steel cabinet (the current design), scores **120**, option B, a rotary magazine behind one hatch, **102**, and option C, a locker bank where each rider charges their own pack in place, **90**, out of a possible 155. A leads B by 15 %, more than the 10 % that would send both to Amish. **Recommendation, proposed, awaiting Amish: carry option A.** The design is not changed by this document. All three options are modelled to detail level D3 in `cad/src/model.py`; the picture below is rendered from those models with `python .kit/options.py`. Method: STANDARDS section 21.



The options

Option A: swap bays in a steel cabinet (current design)

A 1,000 x 500 x 1,500 mm riveted steel cabinet on a channel plinth has four bays in a row, two fitted in the first prototype. Each bay is a steel compartment with a locked door, a cradle and a SwapCell receptacle, and its own charger and CAN channel. The rider taps a card, returns a flat pack to an empty bay and takes a charged one from another, in about 31 s. A 400 W canopy feeds bay 1; a rear plenum and roof hood vent a failed pack away from the street. Risks: R8 fire containment and R9 hot-day charging are at risk; cost is $319 over the target.

Option B: rotary magazine behind one hatch

The same cabinet shell holds a motor-indexed turret with four pockets, each with its own cradle and receptacle, behind a single hatch. The rider puts a flat pack in the hatch pocket and the turret indexes a charged one round to it. One door to guard and seal replaces four, and the pack store is out of reach. Risks: power and data must cross a rotating joint (a slip ring or a flexible lead), not modelled in detail; the turret is a single enclosure for all packs, so a runaway can reach neighbours more easily; one hatch serves one rider at a time; a bought indexing table adds cost (price to assess).

Option C: locker bank, charge in place

Six steel lockers in two columns of three, each with its own door, lock, receptacle and charger. A rider leaves their own SwapCell pack to charge in a locker and returns for it later; there is no pool of packs and no swap. The canopy and MPPT are dropped. Risks: the rider waits hours, not a minute; unvetted packs are stored and charged unattended; six chargers draw about 1.8 kW at once; the top row sits above the comfortable reach range.

First-order numbers

Table 1. Options compared (D3 model figures from `cad/fidelity.json`; masses include the 1,109 kg site pad and anchors).

Quantity	Option A	Option B	Option C
Parts in the D3 model (with fasteners and welds)	537	550	572
Mass of the D3 model (kg)	1,343.8	1,374.1	1,318.3
Priced parts cost, two bays (USD)	1,519 (priced BOM, with anchors and pad)	To assess	To assess
Swap time	31 s	About 31 s plus index time (to assess)	None; the rider waits for a charge
Packs in the pool	3 of 4 bays	3 of 4 pockets	None (riders' own)
Doors	4 bay doors plus a service door	1 hatch plus a service door	6 locker doors plus a service door
Moving parts besides doors	0	Turret and indexer	0
STEM check	Open, one fix made (D1)	Plausible (rotary power transfer)	Within, fire rows open

Scoring

Table 2. Weighted scoring (weights: High UR 3, Medium 2, Low 1; scores 1 to 5; maximum 155).

Criterion	Source	Weight	A	B	C	Reason
Fast swap	UR-001	3	4	4	1	A: 31 s task; B adds the turret index time; C has no swap, the rider waits for their own pack to charge
A charged battery is there	UR-002	3	4	3	2	A: 36 a day ceiling from 4 bays; B: one hatch serves one rider at a time; C: six lockers hold six riders for hours
No fire danger	UR-003	3	3	2	3	A: steel compartment per bay with aerosol and vent; B: packs share one turret enclosure; C: steel locker per pack but unvetted riders' packs
Works outdoors all year	UR-004	3	4	3	3	B has a rotating joint to seal; C has six gasketed doors
Uses the open battery	UR-005	3	5	4	4	B needs power and data across a rotating joint; C charges the pack in place but does not use the swap interface for exchange
Stays up in a storm	UR-006	3	3	3	4	C has no canopy; A and B need the pad (equilibrium factor 1.12)
Hard to rob or tamper with	UR-007	2	4	5	3	B has one hatch to guard; C stores riders' own packs in public lockers
Everyone can use it	UR-008	2	5	4	2	A operable range 0.73 to 1.10 m; C's rows are fixed by the lockers, top row above 1.19 m
Small and easy to power	UR-009	2	4	4	3	C's six chargers draw about 1.8 kW at once, over the 1.5 kW single-circuit target unless limited
No watching me	UR-010	2	4	4	4	No option adds cameras
Quick to mend	UR-011	2	4	2	4	B has an indexer, motor and slip ring behind the hatch
Affordable to build	UR-012	2	3	2	3	A $1,519 priced; B adds a bought indexing table (price to assess); C drops the canopy and MPPT (about $250) but adds four chargers and doors
Some of it from the sun	UR-013	1	3	3	1	C has no canopy
Weighted total			**120**	**102**	**90**	

Decision

Proposed, awaiting Amish: option A, the current design, because:

- It is the only option that delivers the fast swap (UR-001) and a pool of charged packs (UR-002), the two things riders asked for, and it keeps the swap interface the SwapCell project defines (UR-005).
- Its steel compartment per bay is the best of the three on fire containment (UR-003), though that row is still at risk (DKH-STM-001 S11).
- Option B is the nearest alternative (102): worth revisiting if a single hatch matters more than the extra moving parts, or if the propagation trial favours a magazine. Option C is a different product (a charging locker, not a swap station) and fails the two High needs on speed and availability.

The precis (DKH-PRC-001) and every TRL 3 document describe option A. `project.yaml` records `concept_chosen: A (proposed)`. Options B and C stay in the model at D3 as alternatives. The decision record is DKH-DDR-004.